

Theopractical Sheet 3
Partitions and File Systems

1.You are given four files "Extract_1.dd", "Extract_2.dd", "Extract_3.dd" and "Extract_4.dd". These files contain the first three blocks of three USB drives and a portable HDD. You are asked to analyze the partition information (partition types, start, end, sizes) stored in these files. You get 1.25 points for each correctly analyzed file. Please perform the analysis on byte-level, for example using xxd, hexedit or hexdump, as seen in the exercise session.

Ans:{

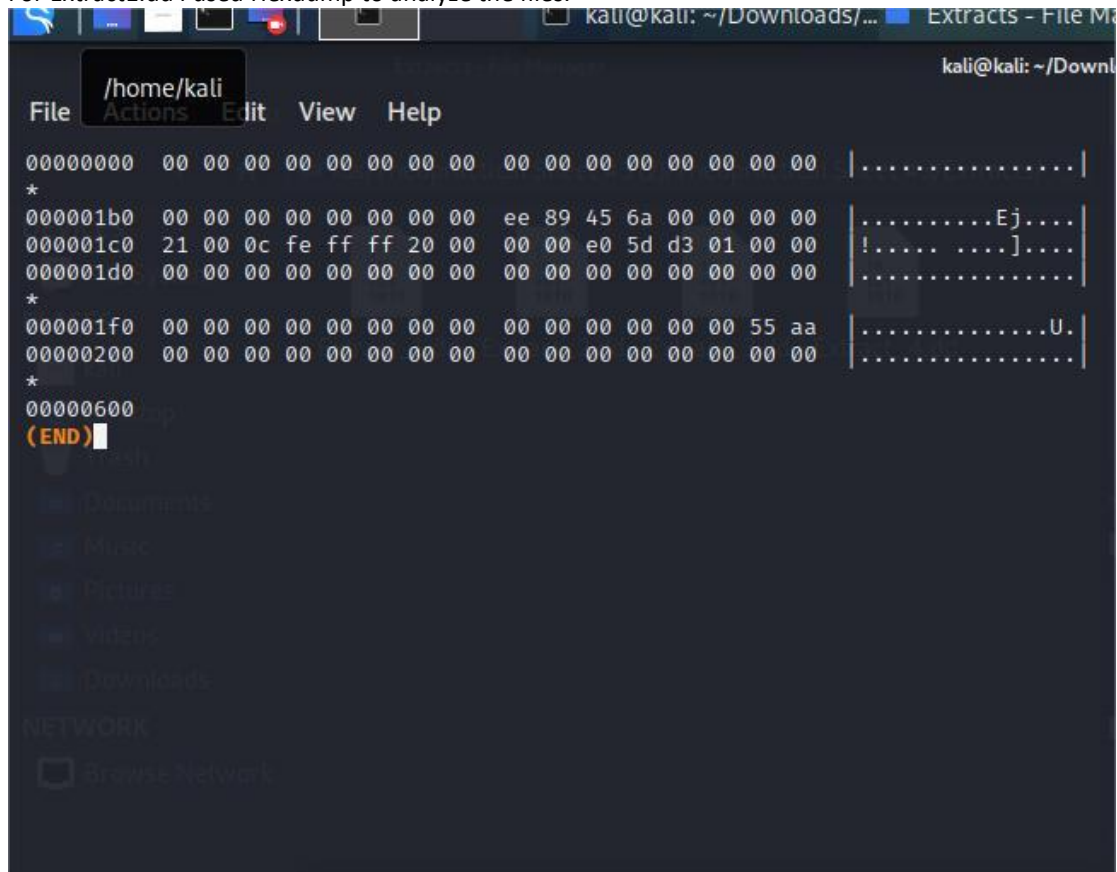
```
"Extract_1.dd": {
  "standard": "MBR",
  "partitions": [
    {
      "partition-type-id": "0x0c",
      "partition-type-name": "FAT32",
      "start-LBA": "0x000020",
      "end-LBA": "0x1d35dff",
      "size-bytes": "15682224128"
    }
  ]
},
"Extract_2.dd": {
  "standard": "MBR",
  "partitions": [
    {
      "partition-type-id": "0x07",
      "partition-type-name": "NTFS",
      "start-LBA": "0x000008",
      "end-LBA": "0x746f67",
      "size-bytes": "3906912256"
    }
  ]
},
"Extract_3.dd": {
  "standard": "GPT",
  "partitions": [
    {
      "partition-type-id": "0xa2",
      "partition-type-name": "EFI",
      "start-LBA": "0x000040",
      "end-LBA": "0x0bce0b3",
      "size-bytes": "6337652736"
    }
  ]
},
"Extract_4.dd": {
  "standard": "MBR",
  "partitions": [
    {
      "partition-type-id": "0x0c",
      "partition-type-name": "FAT32",
      "start-LBA": "0x000008",
```

```

    "end-LBA": "0x35a007",
    "size-bytes": "1799356416"
  }
}
}
}

```

For Extract1.dd I used Hexdump to analyze the files.



Then I analysed the hexdump output and I got that MBR signature 55 AA is at offset 0x1FE and there is no EFI part at offset 0x200.

From the output I got that Partition table starts at 0x1BE and Partition entry found at 0x1C0

Partition type value is 0x0c which is FAT32 and start-LBA = 0x00002000

No of Sectors=0x01d35de0

In order to calculate end LBA I used

end-LBA = start-LBA + num-sectors - 1

size-bytes = num-sectors * 512

For this I used a small python snippet to display the values and calculate end LBA and Size in bytes.

```
# Define values
```

```
start = 0x20
```

```
sectors = 0x1d35de0
```

```
# Compute end LBA
```

```
end = start + sectors - 1
```

```
# Print result in hex
```

```
print("Start LBA:", hex(start))
```

```
print("Number of sectors:", hex(sectors))
print("End LBA:", hex(end))
```

```
(kali@kali)~[~/Downloads/Theopractical Sheet #3 (2)/Theopractical Sheet #3/Extracts]
$ python3
Python 3.8.6 (default, Sep 25 2020, 09:36:53)
[GCC 10.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> # Define values
>>> start = 0x20
>>> sectors = 0x1d35de0
>>>
>>> # Compute end LBA
>>> end = start + sectors - 1
>>>
>>> # Print result in hex
>>> print("Start LBA:", hex(start))
Start LBA: 0x20
>>> print("Number of sectors:", hex(sectors))
Number of sectors: 0x1d35de0
>>> print("End LBA:", hex(end))
End LBA: 0x1d35dff
>>>
```

Similarly I followed the same steps for Extract2.dd, Extract3.dd and Extract4.dd where I analysed the hexdump output to figure out if its MBR or GPT and figure out the start LBA and number of sectors and partition type based on the value. Then use the snippet to calculate the end LBA and size in bytes.

For Extract2.dd

```
File Actions Edit View Help
00000010 b f 1b 06 50 57 b9 e5 01 f3 a4 cb bd be 07 b1 04 ... PW.....
00000020 38 6e 00 7c 09 75 13 83 c5 10 e2 f4 cd 18 8b f5 8n. | .u.....
00000030 83 c6 10 49 74 19 38 2c 74 f6 a0 b5 07 b4 07 8b ... It.8,t.....
00000040 f0 ac 3c 00 74 fc bb 07 00 b4 0e cd 10 eb f2 88 ..<.t.....
00000050 4e 10 e8 46 00 73 2a fe 46 10 80 7e 04 0b 74 0b N..F.s*.F..~.t.
00000060 80 7e 04 0c 74 05 a0 b6 07 75 d2 80 46 02 06 83 ~..t...u..F...
00000070 46 08 06 83 56 0a 00 e8 21 00 73 05 a0 b6 07 eb F...V...!.s....
00000080 bc 81 3e fe 7d 55 aa 74 0b 80 7e 10 00 74 c8 a0 ..>}.U.t..~..t..
00000090 b7 07 eb a9 8b fc 1e 57 8b f5 cb bf 05 00 8a 56 .....W.....V
000000a0 00 b4 08 cd 13 72 23 8a c1 24 3f 98 8a de 8a fc .....r#..$?.....
000000b0 43 f7 e3 8b d1 86 d6 b1 06 d2 ee 42 f7 e2 39 56 C.....B..9V
000000c0 0a 77 23 72 05 39 46 08 73 1c b8 01 02 bb 00 7c .w#r.9F.s..... |
000000d0 8b 4e 02 8b 56 00 cd 13 73 51 4f 74 4e 32 e4 8a .N..V...sQOtN2..
000000e0 56 00 cd 13 eb e4 8a 56 00 60 bb aa 55 b4 41 cd V.....V..`..U.A.
000000f0 13 72 36 81 fb 55 aa 75 30 f6 c1 01 74 2b 61 60 .r6..U.u0...t+a`
00000100 6a 00 6a 00 ff 76 0a ff 76 08 6a 00 68 00 7c 6a j.j..v..v.j.h.|j
00000110 01 6a 10 b4 42 8b f4 cd 13 61 61 73 0e 4f 74 0b .j..B....aas.Ot.
00000120 32 e4 8a 56 00 cd 13 eb d6 61 f9 c3 49 6e 76 61 2..V.....a..Inva
00000130 6c 69 64 20 70 61 72 74 69 74 69 6f 6e 20 74 61 lid partition ta
00000140 62 6c 65 00 45 72 72 6f 72 20 6c 6f 61 64 69 6e ble.Error loadin
00000150 67 20 6f 70 65 72 61 74 69 6e 67 20 73 79 73 74 g operating syst
00000160 65 6d 00 4d 69 73 73 69 6e 67 20 6f 70 65 72 61 em.Missing opera
00000170 74 69 6e 67 20 73 79 73 74 65 6d 00 00 00 00 00 ting system.....
00000180 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
*
000001b0 00 00 00 00 00 2c 44 63 00 a2 35 ff 00 00 00 20 |.....,Dc..5....
000001c0 21 00 07 fe ff ff 00 08 00 00 00 60 6f 74 00 00 !.....`ot..
000001d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
*
000001f0 00 00 00 00 00 00 00 00 00 00 00 00 55 aa |.....U.
:|
```


For Extract3.dd

```
kali@kali: ~/Downloads/... kali@kali: ~/Downloads/... Extracts - File ...
kali@kali: ~/Downloads/Theopractical Sheet #3 (2)/T

File Actions Edit View Help

00000000 eb 63 90 90 90 90 90 90 90 90 90 90 90 .c..... Extracts
00000010 90 90 90 90 90 90 90 90 90 90 90 90 ..I$
00000020 00 52 be 1b 7c 31 c0 cd 13 46 8a 0c 84 c9 75 10 .R..|1...F...u.
00000030 be 39 7c e8 74 01 e9 3d 01 46 6c 6f 70 70 79 00 .9|.t..=.Floppy.
00000040 bb 00 70 8e c3 31 db b8 01 02 b5 00 b6 00 cd 13 ..p..1.....
00000050 72 d3 b6 01 b5 4f e9 98 00 00 00 80 00 00 00 00 r...0.....
00000060 00 00 00 00 ff fa eb 05 f6 c2 80 74 05 f6 c2 70 .....t...p
00000070 74 02 b2 80 ea 79 7c 00 00 31 c0 8e d8 8e d0 bc t....y|..1.....
00000080 00 20 fb a0 64 7c 3c ff 74 02 88 c2 52 be 80 7d ..d|<.t...R..}
00000090 e8 17 01 be bf 7d b4 41 bb aa 55 cd 13 5a 52 72 ....}.A..U..ZRR
000000a0 3d 81 fb 55 aa 75 37 83 e1 01 74 32 31 c0 89 44 =..U.u7...t21..D
000000b0 04 40 88 44 ff 89 44 02 c7 04 10 00 66 8b 1e b0 [.0.D...D.....f...
000000c0 7d 66 89 5c 08 66 8b 1e b4 7d 66 89 5c 0c c7 44 [f...\f...}f.\..D
000000d0 06 00 70 b4 c2 cd 13 72 05 bb 00 70 eb 76 b4 08 ..p.B..r...p.v..
000000e0 cd 13 73 0d 5a 84 d2 0f 83 36 ff be 8b 7d e9 82 ..s.Z....6...}..
000000f0 00 66 0f b6 c6 88 64 ff 40 66 89 44 04 0f b6 d1 .f....d.f.D....
00000100 c1 e2 02 88 ee 88 f4 40 89 44 08 0f b6 c2 c0 e8 .....0.D.....
00000110 02 66 89 04 66 a1 b4 7d 66 09 c0 75 4e 66 a1 b0 .f...f...}f..uNf..
00000120 7d 66 31 d2 66 f7 34 88 d1 31 d2 66 f7 74 04 3b [f1.f.4...1.f.t.;
00000130 44 08 7d 37 fe c1 88 c5 30 c0 c1 e8 02 08 c1 88 D.|7....0.....
00000140 d0 5a 88 6b bb 00 70 8e c3 31 db b8 01 02 cd 13 .Z....p..1.....
00000150 72 1e 8c c3 60 1e b9 00 01 8e db 31 f6 bf 00 80 r...`.....1....
00000160 8e c6 fc f3 a5 1f 61 ff 26 5a 7c be 86 7d eb 03 .....a.8Z|..}..
00000170 be 95 7d e8 34 00 be 9a 7d e8 2e 00 cd 18 eb fe ...4....}...
00000180 47 52 55 42 20 00 47 65 6f 6d 00 48 61 72 64 20 GRUB .Geom.Hard
00000190 44 69 73 6b 00 52 65 61 64 00 20 45 72 72 6f 72 Disk.Read. Error
000001a0 0d 0a 00 bb 01 00 b4 0e cd 10 ac 3c 00 75 f4 c3 .....<.u...
000001b0 f8 0f 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
000001c0 02 00 ee ff ff ff 01 00 00 00 ff 3f ca 01 80 00 .....?....
000001d0 01 00 00 00 01 00 00 00 00 00 01 00 00 00 00 .....
000001e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
000001f0 00 00 00 00 00 00 00 00 00 00 00 00 55 aa .....U.
00000200 45 46 49 20 50 41 52 54 00 00 01 00 5c 00 00 00 EFI PART....\...
00000210 b6 49 6d 36 00 00 00 00 01 00 00 00 00 00 00 .Im6.....
00000220 ff 3f ca 01 00 00 00 00 40 00 00 00 00 00 00 .?......0.....
00000230 c0 3f ca 01 00 00 00 00 a3 6c ef b0 d1 60 21 4a .?...l...!J
00000240 98 d9 17 c3 85 a8 b6 f8 02 00 00 00 00 00 00 .....
00000250 f8 00 00 00 80 00 00 00 ea d2 f0 13 00 00 00 .....
00000260 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

*
00000400 a2 a0 d0 eb e5 b9 33 44 87 c0 68 b6 b7 26 99 c7 .....3D...h...6..
00000410 a3 6c ef b0 d1 60 21 4a 98 d8 17 c3 85 a8 b6 f8 .l...`!J.
00000420 40 00 00 00 00 00 00 00 b3 e0 bc 00 00 00 00 00 @.....
00000430 01 00 00 00 00 00 00 10 49 00 53 00 4f 00 39 00 .....I.S.O.9.
00000440 36 00 36 00 30 00 00 00 00 00 00 00 00 00 00 6.6.0.....
```

```
(kali@kali)~[~/Downloads/Theopractical Sheet #3 (2)/Theopractical Sheet #3/Extracts]
$ python3
Python 3.8.6 (default, Sep 25 2020, 09:36:53)
[GCC 10.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> start = 0x40
>>> end = 0xbce0b3
>>> sectors = end - start + 1
>>> size_bytes = sectors * 512
>>>
>>> print("Start LBA:", hex(start))
Start LBA: 0x40
>>> print("End LBA:", hex(end))
End LBA: 0xbce0b3
>>> print("Sectors:", sectors)
Sectors: 12378228
>>> print("Size (bytes):", size_bytes)
Size (bytes): 6337652736
>>>
```

For Extract4.dd

```
(kali@kali)~[~/Downloads/Theopractical Sheet #3 (2)/Theopractical Sheet #3/Extracts]
$ python3
Python 3.8.6 (default, Sep 25 2020, 09:36:53)
[GCC 10.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> start = 0x08
>>> sectors = 0x35a000
>>> end = start + sectors - 1
>>> size_bytes = sectors * 512
>>>
>>> print("Start LBA:", hex(start))
Start LBA: 0x8
>>> print("End LBA:", hex(end))
End LBA: 0x35a007
>>> print("Size (bytes):", size_bytes)
Size (bytes): 1799356416
>>>
```

2. You are given two files "4_Broken_File_1.dat" and "4_Broken_File_2.dat" that contain an image (= picture) file with a flag each. Use command-line tools to carve the images. Open the file and give the flag. Document the commands that you have used in your proof-of-work file. Give the flag including FLAG{...} as a string. Additionally, provide the offset at which the image starts in bytes as a decimal number (without separators). You likely needed this as a parameter during carving

Ans:{

```
"question_2": {
  "4_Broken_File_1.dat": {
    "flag": "FLAG{4n6_RH1N0_F1}",
    "offset": "462"
  },
  "4_Broken_File_2.dat": {
    "flag": "FLAG{NEBELSCHWADENFROH}",
    "offset": "84"
  }
}
```

For both the dat files I followed the same steps starting with analysing the file in hexdump and using that output to figure out the image header

For 4_Broken_File_1.dat


```

00000000 92 91 53 42 54 31 56 17 11 81 81 81 19 19 10 .. SBT1V.....
00000010 10 10 15 65 24 54 26 54 25 45 41 54 54 15 64 65 ... e$T6TxEATT.de
00000020 41 54 65 45 14 65 46 51 46 54 16 54 65 41 65 45 ATeE..eFQFT.TeAeE
00000030 64 16 54 65 45 61 da fe da bc d6 16 16 15 15 15 d.TeEa.....
00000040 15 15 15 15 15 11 77 71 71 71 77 17 17 71 71 .....wqqqw..qq
00000050 77 17 17 71 71 77 17 17 bd eb de bd eb de af w..qqqw.....
00000060 af af af af af af af af e4 11 41 44 14 41 44 41 .....AD.ADA
00000070 44 14 55 16 17 17 16 16 15 14 41 34 15 17 81 99 D.U.....A4....
00000080 19 01 01 01 00 10 10 10 10 10 10 10 10 10 01 .....
00000090 18 17 18 61 87 61 56 15 14 54 51 65 16 71 18 81 ... a..aV..TQe.q..
000000a0 01 18 18 71 89 79 17 98 71 98 71 98 67 15 74 14 ... q..y..q.q.g.t.
000000b0 14 53 41 16 57 86 18 61 89 71 87 19 87 18 91 78 .SA.W..a.q.....x
000000c0 91 78 71 89 79 81 79 87 18 71 75 16 41 65 46 15 .xq.y.y..qu.AeF.
000000d0 45 61 46 53 14 31 56 16 57 16 18 79 80 98 10 78 EaFS.1V.W..y...x
000000e0 71 57 14 56 14 56 41 56 41 54 16 54 16 54 16 54 qW.V.VAVAT.T.T.T
000000f0 65 14 65 14 16 54 16 54 15 64 16 54 15 64 15 64 e.e..T.T.d.T.d.d
00000100 16 54 15 61 46 51 46 51 45 61 46 54 16 51 46 51 .T.aFQFQeAFT.QFQ
00000110 41 65 41 65 14 65 14 51 64 65 14 56 14 16 54 61 AeAe..e.Qde.V..Ta
00000120 54 16 54 16 51 46 51 45 61 41 65 41 65 41 65 41 T.T.QFQeAeAeAeA
00000130 65 41 56 1f af ad af ef ef ef ed ed ed ed ed ed eAV.....
00000140 ed ed ed de de 51 51 51 51 51 51 51 54 14 13 13 15 .....QQQQQT....
00000150 16 17 11 81 91 71 61 51 41 31 45 16 17 18 18 19 .....qaQA1E.....
00000160 19 18 71 61 15 14 41 31 41 51 65 76 15 76 15 67 ..qa..A1Aqev.v.g
00000170 51 67 56 71 56 71 57 65 17 65 17 65 76 15 76 51 QgVqVqWe.e.ev.vQ
00000180 76 57 61 57 65 17 65 17 65 67 15 76 51 76 51 76 vWaWe.e.eg.vQvQv
00000190 57 61 56 15 76 15 87 16 17 86 87 65 17 64 51 64 WaV.v.....e.dQd
000001a0 35 43 15 46 51 47 65 17 66 18 76 61 56 74 51 45 5C.FQGe.f.vaVtQE
000001b0 13 56 41 54 56 15 17 81 57 61 57 61 56 17 5d da .VATV ...WaWaV.].
000001c0 da da da da da dd ad ad ad ad ad a0 00 ff d8 .....
000001d0 ff e0 00 10 4a 46 49 46 00 01 01 01 00 60 00 60 .....JFIF.....
000001e0 00 00 ff db 00 43 00 03 02 02 03 02 02 03 03 03 .....C.....
000001f0 03 04 03 03 04 05 08 05 05 04 04 05 0a 07 07 06 .....
00000200 08 0c 0a 0c 0c 0b 0a 0b 0b 0d 0e 12 10 0d 0e 11 .....
00000210 0e 0b 0b 10 16 10 11 13 14 15 15 15 0c 0f 17 18 .....
00000220 16 14 18 12 14 15 14 ff db 00 43 01 03 04 04 05 .....C.....
00000230 04 05 09 05 05 09 14 0d 0b 0d 14 14 14 14 14 14 .....
00000240 14 14 14 14 14 14 14 14 14 14 14 14 14 14 14 .....
*
00000260 14 14 14 14 14 14 14 14 14 14 14 14 ff c2 00 11 .....
00000270 08 04 63 03 1a 03 01 22 00 02 11 01 03 11 01 ff ..C.....
00000280 c4 00 1d 00 00 01 05 01 01 01 01 00 00 00 00 00 .....
00000290 00 00 00 00 00 01 02 03 04 05 06 07 08 09 ff c4 .....
000002a0 00 1a 01 00 03 01 01 01 01 00 00 00 00 00 00 00 .....
000002b0 00 00 00 00 01 02 03 04 05 06 ff da 00 0c 03 01 .....
000002c0 00 02 10 03 10 00 00 01 f5 21 4f 96 f7 41 16 44 .....I.O..A.D

```

So from the output I found these

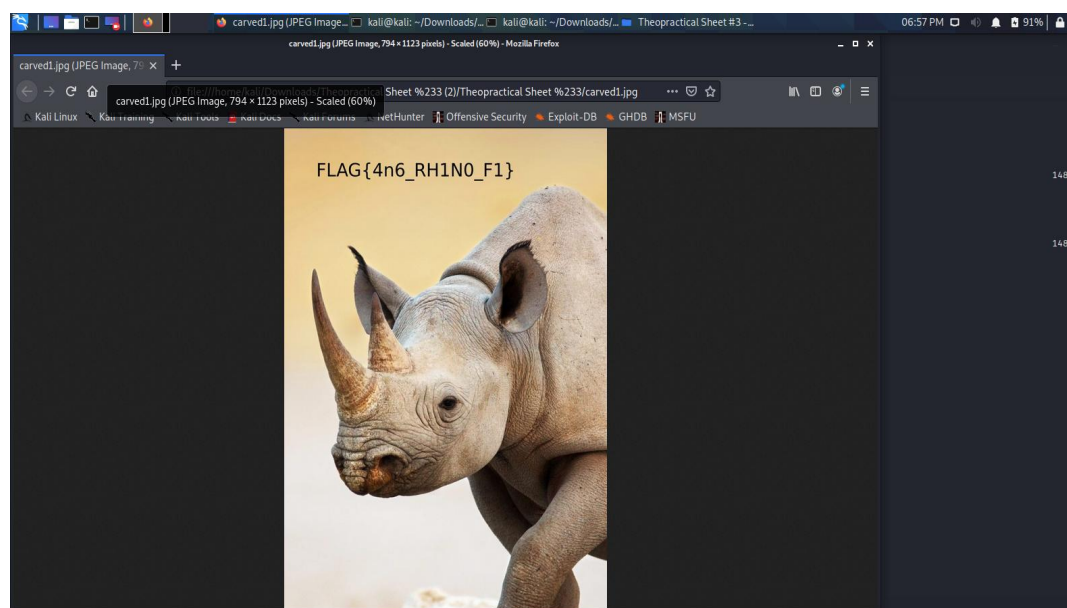
000001cc a0 00 ff d8

000001d0 ff e0 00 10 4a 46 49 46

From this we know image starts from ff and ff d8 ff e0 are image headers and 4a 46 49 46 confirms JPEG Format.

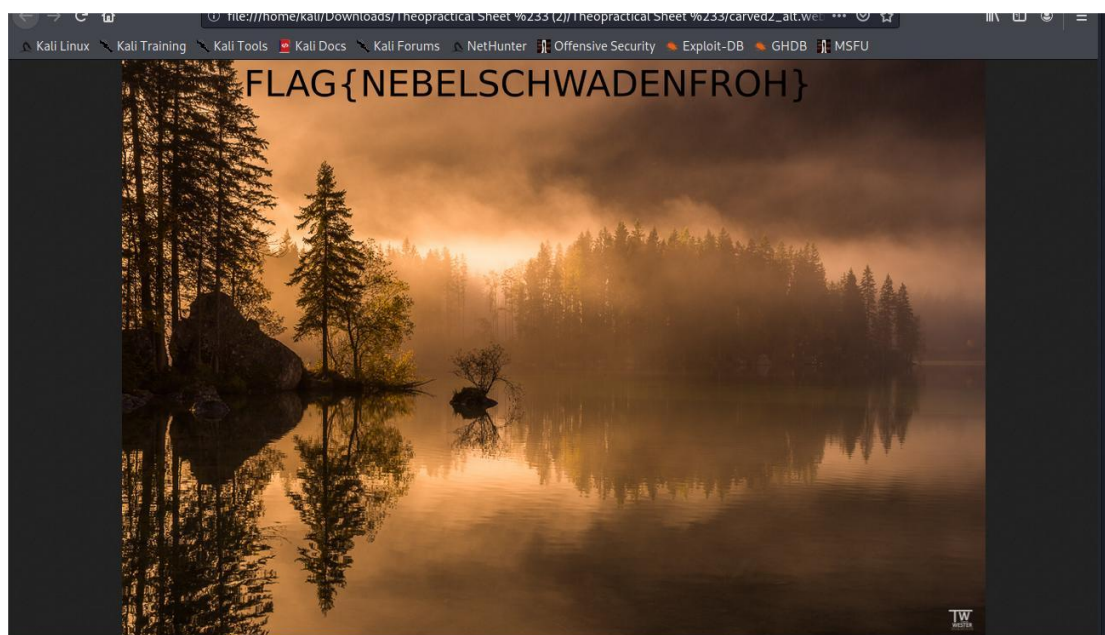
The image starts at offset 0x1ce and basically in decimal it means 462.

I used dd command in terminal to extract the image from the offset 462.I got a rhino picture and a text FLAG{4n6_RH1N0_F1}.



Similarity for "4_Broken_File_2.dat" I got a .webp at offset 84 with text FLAG{NEBELSCHWADENFROH}

```
00000000 77 36 62 52 51 55 35 36 63 63 63 66 35 25 52 52 w6bRQU56cccf5%RR
00000010 52 51 51 51 51 72 82 39 39 37 62 52 52 52 52 df RQQQr.997bRRRR.
00000020 cc dd aa bb 55 62 62 62 62 62 62 52 52 52 52 ....UbbbbbbRRRRR
00000030 55 25 15 15 15 15 15 15 15 15 15 15 15 15 61 U%.....QQa
00000040 71 71 78 88 87 72 62 62 27 18 76 76 85 56 77 65 qqx..rbb'.vv.Vwe
00000050 78 76 77 bb 52 49 46 46 2e c1 0b 00 57 45 42 50 xvw.RIFF....WEBP
00000060 56 50 38 4c 22 c1 0b 00 2f ff 83 aa 00 0d 75 21 VP8L" .../.....u!
00000070 a2 ff c1 32 90 b6 4d db f9 57 fd 8b 88 88 09 80 ... 2..M..W.....
00000080 5e 98 6a 83 99 6e 4f b0 2e f4 72 d8 b6 6d d9 64 ^.j..n0...r..m.d
00000090 27 fa d2 ae eb be ef c7 5e 7f df 72 ef ae 76 5f '.....^..r..v_
000000a0 bd dc 35 ee 09 09 10 dc 1d c6 f9 46 19 c6 67 60 ..5.....F..g_
000000b0 06 f7 24 40 dc 3d 2b cb 5d db dd bb dc e5 75 79 ..$@*+=.]......uy
000000c0 e4 96 6b ab 7a 9e 6a 3e 5f d7 36 ab b6 6d 3b ce ..k.z.j>_.6..m;.
000000d0 17 51 a1 41 e7 81 93 e7 5a 73 c1 e6 7d 8e 8e 28 .Q.A....Zs..}..(
000000e0 95 66 be f3 1f f5 95 2f d9 69 3b 95 92 0e 6d 5e .f...../..i;...m^
000000f0 38 79 ce c1 63 74 6c 58 21 c2 bb b5 4d aa 6d 49 8y..ctlX!...M.mI
00000100 92 f4 89 28 4c 5a bc f1 ec c3 e7 18 b9 99 39 05 ... (LZ.....9.
00000110 66 46 64 71 33 33 5c 42 df 0e 77 ff 64 66 66 2c fFdq33\B..w.dff,
00000120 c8 cc 40 67 73 c3 63 07 37 2f 9c a4 20 12 14 60 ..@gs.c.7/.. ..
00000130 db 8e db 36 9d 24 4d 1c 2f 30 fe fb 0b 3e 80 8f ... 6.$M./0...>..
00000140 85 20 95 f1 70 6d 9b 22 49 92 a4 4f e9 79 5e 11 . ..pm."I..O.y^.
00000150 35 33 87 88 c8 48 28 6e 9c d9 99 e5 dd 8f 19 7e 53 ...H(n.....~
00000160 3b f3 e2 ec 0e f4 34 17 24 06 ba bb 99 a9 8a bc ;.....4.$.....
00000170 cf 11 79 5f b5 6a 0f b6 2d 39 92 24 49 d0 25 22 ..y_.j..-9.$I.%"
00000180 11 35 33 f7 48 50 59 d5 68 30 46 fb 5f 0c c6 33 .53.HPY.h0F...3
00000190 3d 85 2b 33 22 dc cd 54 45 88 c8 bb 80 49 8e 24 =.+3" ..TE...I.$
000001a0 49 92 f4 11 b1 aa 99 7b 44 66 16 e8 1e b8 87 f1 I.....{Df.....
000001b0 fb 3f c7 fd c6 77 8b c1 a0 ee ae 4a 10 ee 6e aa .? ...w.....J..n.
000001c0 4c 14 14 68 db a9 dd 46 b3 a6 33 6d 9a a4 fc 74 L..h ... F..3m ...t
000001d0 11 20 84 10 bc d7 b5 bc 6d db ad da b6 1b c9 1b . ....m.....
000001e0 7b 5b 1f 73 ed bd 0f 89 65 d9 96 64 87 99 82 31 {[.s....e...d...1
000001f0 21 18 93 99 af 4a 2e 66 a6 4f 5c f5 07 98 99 99 !....J.f.0\....
00000200 19 92 19 23 03 1c 60 94 cc e2 03 7b af 35 47 6b ...#..`.....{.5Gk
00000210 57 6f ef d3 c6 3a e5 61 db b6 65 7b db 4c 5b 8f Wo ... :.a..e{.L[.
00000220 e3 bc ef 47 2c b3 03 0d 96 99 86 79 e6 ff f3 7c ...G,.....y ...|
00000230 fc bd 5c ae 9d c4 b1 0c 92 9e fb 3e f6 c3 a3 6d ..\.....> ...m
00000240 db ae 6d b7 91 55 fb 98 fb dc 8b 0b 12 20 65 c2 ..m..U.....e.
00000250 47 66 3e 6f fe ff 33 d2 fb 8c 08 b9 08 49 a4 0c Gf>o..3.....I..
00000260 2d 80 eb ce d9 73 f4 a0 00 db 4a 25 49 da 98 99 .....s....J%I...
00000270 9d ee e9 0f 22 20 2a 22 9a 66 56 87 c7 6d 9f 55 ...." *.fv..m.U
00000280 db 72 63 79 7e 01 63 cc b9 78 e3 d9 fb 30 25 67 .rcy~.c..x ...0%g
00000290 2a 95 52 8a 2c 83 64 c6 72 b9 f8 42 d5 65 6a a6 *.R.,.d.r..B.ej.
000002a0 db cc dd 97 99 f9 16 bb c8 5c 86 32 c9 96 65 81 .....\.2..e.
000002b0 25 a5 a4 54 32 1d e6 7d 36 2f 9c 73 8e 11 11 cf %..T2..}6/.s....
000002c0 9a 63 ec ac 6e 6f db 36 29 92 6d 49 d2 23 22 6a .c..no.6).mI.#"j
```



3.Which of the following conditions need to be fulfilled in order to be able to carve a file successfully? Select all that apply and copy them as strings into your answer array.

- "The entire metadata entry must still be present."
- "The header, footer and most of the blocks of the file were not overwritten with other data."
- "The file needs to be present in the operating system's cache."
- "The Gutmann scheme was successfully executed"

```
Ans : "question_3": [  
  "The header, footer and most of the blocks of the file were not overwritten with other data."  
]
```

4.According to Carrier's Data Categories, what does the "File System" category tell you?
"Data unit" is an abstract term referring to blocks, sectors, etc., depending on the file system.

```
Ans: "question_4": [  
  "The location of a data unit."  
]
```

5.Why does it make sense to use Carrier's Data Categories in Forensic Science? Select all reasons that apply.

```
Ans: "question_5": [  
  "They provide a general interface for forensic tools, independently from concrete file systems.",  
  "They can be helpful to prove a trace's integrity.",  
  "They allow to compare file systems abstractly."  
]
```

In digital forensics Carrier's data categories are useful because they provide a simple yet standardised method for analysing and categorising data regardless of the file system that is being used.They can also be used to interpret and trace data and abstraction provides consistency while comparing different file sytem formats.

Putting everything together in one json:

```
{  
  "name": "Thota Vijay Kumar",  
  "matriculation": "7056597",  
  "question_1": {  
    "Extract_1.dd": {  
      "standard": "MBR",  
      "partitions": [  
        {  
          "partition-type-id": "0x0c",  
          "partition-type-name": "FAT32",  
          "start-LBA": "0x000020",  
          "end-LBA": "0x1d35dff",  
        }  
      ]  
    }  
  }  
}
```



```

    "size-bytes": "15682224128"
  }
]
},
"Extract_2.dd": {
  "standard": "MBR",
  "partitions": [
    {
      "partition-type-id": "0x07",
      "partition-type-name": "NTFS",
      "start-LBA": "0x000008",
      "end-LBA": "0x746f67",
      "size-bytes": "3906912256"
    }
  ]
},
"Extract_3.dd": {
  "standard": "GPT",
  "partitions": [
    {
      "partition-type-id": "0xa2",
      "partition-type-name": "EFI",
      "start-LBA": "0x000040",
      "end-LBA": "0x0bce0b3",
      "size-bytes": "6337652736"
    }
  ]
},
"Extract_4.dd": {
  "standard": "MBR",
  "partitions": [
    {
      "partition-type-id": "0x0c",
      "partition-type-name": "FAT32",
      "start-LBA": "0x000008",
      "end-LBA": "0x35a007",
      "size-bytes": "1799356416"
    }
  ]
}
},
"question_2": {
  "4_Broken_File_1.dat": {
    "flag": "FLAG{4n6_RH1N0_F1}",
    "offset": "462"
  },
  "4_Broken_File_2.dat": {
    "flag": "FLAG{NEBELSCHWADENFROH}",
    "offset": "84"
  }
},
"question_3": [
  "The header, footer and most of the blocks of the file were not overwritten with other data."
],
"question_4": [
  "The location of a data unit."
],

```

```
"question_5": [  
  "They provide a general interface for forensic tools, independently from concrete file systems.",  
  "They can be helpful to prove a trace's integrity.",  
  "They allow to compare file systems abstractly."  
]  
}
```